WILBERT ANDREW YONATHAN

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EDUCATION

Xiamen University Malaysia Sep 2023 – Sep 2027 (Expected)
Bachelor of Engineering in Artificial Intelligence | CGPA: 3.77/4.00

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Javascript, SQL
Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, LightGBM, YOLOv8, Object Detection, Image Classification, Visual Anomaly Detection
Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face Transformers, Amazon Bedrock
Data & Databases: NumPy, pandas, SQLite, Supabase (PostgreSQL)
Web & APIs: FastAPI, Flask, React (React.js), Gradio, Streamlit, REST APIs
Cloud & Deployment: AWS (Lightsail, EC2, Lambda, CloudWatch, Budgets), Docker, Docker Compose, Linux (Ubuntu), Git, Hugging Face Spaces, Render, Vercel
Automation & Integration: n8n, Webhooks, Resend, Formspree
Spoken Languages: English, Mandarin, Indonesian, Malay, Hokkien

PROJECT EXPERIENCE

PaddyLiteX: Paddy Disease Detection Apr 2026 – Present
- Developed a DenseLiteX-SPPF-C2PSA YOLOv8n detector for nine paddy classes, achieving 81.67% mAP@0.5 (+0.88 percentage points) with 4.24% fewer parameters than YOLOv8n.
- Compared with the initial YOLOv8n-DenseNet121 proposal, reduced parameters by 68.01% and GFLOPs by 72.59%; conducted ablation studies and deployed a three-model comparison demo.

DefectRAG: Visual Anomaly Inspection and RAG Aug 2026 – Oct 2026
- Built and evaluated a frozen DINOv2 anomaly-inspection pipeline across 12 VisA products, achieving 95.13% mean image AUROC in retrospective evaluation.
- Integrated BGE embeddings, Qwen, LangChain and LangGraph for cited RAG; built a React.js/FastAPI dashboard, packaged locally with Docker and deployed a Hugging Face ZeroGPU demo.
- Built corrective-action tracking using Supabase and n8n workflows for event validation, supervisor notifications, and delivery receipts; deployed n8n on AWS Lightsail using Docker Compose and a Caddy reverse proxy with HTTPS.

MarineLite: Marine Organism Detection Jun 2026 – Jul 2026
- Integrated GhostConv and SimSPPF into DU-MobileYOLO; increased mAP@0.5 from 74.23% to 75.36% on URPC2020 while reducing parameters from 4.70M to 4.37M.
- Conducted ablation studies and deployed a web demo with detection overlays and model comparisons.

dPickleball: Reinforcement Learning Agent May 2026 – Jul 2026
- Trained Proximal Policy Optimization (PPO) agents using Unity ML-Agents, curriculum learning, reward shaping and league self-play.
- Contributed policies and round-robin evaluation to a six-member team that placed 2nd among 13 teams in the final class competition.

WORK EXPERIENCE

Student Registration & Operations Assistant, Xiamen University Malaysia Apr 2025
- Supported registration logistics and directed students as part of a team serving 800+ students.
- Organized university assets and materials for event operations and proper storage.

LEADERSHIP AND ACTIVITIES

Head of Photography & Videography, Love in Action Feb 2026 – Mar 2026
Coordinated media coverage and edited outreach materials.

Head of General Affairs, XMUM AI Club Apr 2025 – Oct 2025
Coordinated logistics and external partners for a national-level AI event.

Assistant of General Affairs, AIESEC XMUM Sep 2023 – Sep 2025
Coordinated room bookings and venue setups for weekly meetings.

AWARDS

- Dean's List: Three consecutive semesters.
- XMUM Merit Scholarship: Four-year undergraduate programme (2023–2027).

REFERENCES

Shaidah Binti Jusoh
Assistant Professor | Thesis Supervisor
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Goh Sim Kuan
Assistant Professor | Internship Coordinator
simkuan.goh@xmu.edu.my | 012-7660576